International Journal of Biological Macromolecules 186 (2021) 535–543

Contents lists available at ScienceDirect

International Journal of Biological Macromolecules

journal homepage: www.elsevier.com/locate/ijbiomac

Five distinct fucan sulfates from sea cucumber *Pattalus mollis*: Purification, structural characterization and anticoagulant activities

Yan Ma [a,b], Na Gao [b], Zhichuang Zuo [b], Shanni Li [c], Wenqi Zheng [b], Xiang Shi [b], Qipei Liu [b], Ting Ma [a,b], Ronghua Yin [b,*], Xian Li [a,*], Jinhua Zhao [b,*]

[a] School of Pharmaceutical Sciences and Yunnan Key Laboratory of Pharmacology for Natural Products, Kunming Medical University, Kunming 650500, China
[b] School of Pharmaceutical Sciences, South-Central University for Nationalities, Wuhan 430074, China
[c] State Key Laboratory of Phytochemistry and Plant Resources in West China, Kunming Institute of Botany, Chinese Academy of Sciences, Kunming 650201, China

ARTICLE INFO

Keywords:
Sea cucumber
Fucan sulfate
Chemical structure
Anticoagulant
Pattalus mollis

ABSTRACT

Fucan sulfates from echinoderm possess characteristic structures and various biological activities. Herein, comprehensive methods including enzymolysis, ion-exchange chromatography and size exclusion chromatography lead to the purification of five fucan sulfates (FSI, FSII, FSIII, FSIV, FSV) from the sea cucumber *Pattalus mollis*. Chemical composition analysis showed that they were all composed of L-fucose. Their sulfate content was determined by a conductimetric method. The molecular weight (Mw) of FSI, FSII, FSIII, FSIV and FSV were measured as 238.3 kDa, 81.0 kDa, 82.0 kDa, 23.2 kDa and 6.12 kDa, respectively. Detailed NMR spectroscopic analysis revealed that the structural sequence of FSI and FSII was →3)-L-Fuc_S-α(1→, where Fuc_S were Fuc_{2S4S} (10%), Fuc_{2S} (44%), Fuc_{0S} (10%), Fuc_{4S} (36%), that of FSIII was →4)-L-Fuc_{2S}-(α1 → 4)-L-Fuc_{2S}-(α1 → 4)-L-$Fuc_{0S/3S}$-(α1→, where Fuc_{0S} and Fuc_{3S} were in equal molar, and that FSIV was →4)-L-Fuc_{2S3S}-(α1 → 4)-L-Fuc_{2S3S}-(α1 → 4)-L-Fuc_{2S}-(α1→4)-L-Fuc_{2S}-(α1 → 4)-L-Fuc_{2S}-(α1 → 4)-L-Fuc_{2S}-(α1 → . This is the first report that such a diversity of fucan sulfates were obtained from the same sea cucumber species. Biological activity showed that FSI, FSII, FSIII and FSIV exhibited potent anticoagulant by prolonging the APTT. Among them, FSII, FSIII and FSIV showed the similar potency, while FSI owned the strongest. Structure-activity relationships analysis showed that molecular weight and sulfation degree should be the crucial factors for the activity.

1. Introduction

Sea cucumbers (Echinodermata, Holothuroidea) are important human food sources for their nutrition and medicinal benefits. The processed body wall is mainly exported from producing countries to Asia, especially Chinese seafood markets, where it is popular and of high value [1]. Studies have shown sea cucumber contains protein, collagen fibers, amino acids, and bioactive components such as glycosides and sulfated polysaccharides. Sulfated polysaccharides such as fucan sulfate are endowed with various biological activities, including anticoagulation and anti-thrombosis [2], hematopoietic stimulation [3], anti-cancer [4], hepatic inflammatory response and insulin resistance modulation [5].

Fucan sulfate is mainly composed of fucose and sulfate groups that is mostly found in brown algae and invertebrates. The fucan sulfates isolated from brown algae was also called as fucoidan. In contrast with the apparently complex and heterogeneous algal fucoidans, the fucan sulfate from invertebrates (FS) such as sea cucumber usually possesses more regular structures. Most of FSs are linear polysaccharides linked by α(1 → 3), α(1 → 4) or α(1 → 2) glycosidic bonds in repeating units, and the α-L-fucose residues sulfated at O-2, O-3 and/or O-4 positions. For example, FSs from *Holothuria fuscopunctata* [6], *Stichopus herrmanni* [7], *Stichopus horrens* [8] are highly regular fucan sulfates with monosaccharide repeating units. FSs from *Acaudina molpadioides* [9], *Holothuria polii* [3], *Holothuria tubulosa* [10], *Isostichopus badionotus* [2], *Thelenota ananas* [11] are all linear polysaccharides consisting of α(1 → 3) linked tetrafucose repeating units, but with different sulfation patterns. FS from *Holothuria albiventer* is composed of regular α(1 → 3) linked hexasaccharide repeating units [12]. Additionally, some FSs were discovered to bear the mono-fucose and di-fucose side chain that linked to the central core. FSs from *Holothuria coluber* [12], *Apostichopus japonicas* [13], *Pattalus mollis* [14] are branched polysaccharides, in which the repeating structural unit of AjFS has a disaccharide side chain, and the repeating structural unit of HcFS and PmFS are branched by a mono-

* Corresponding authors.
E-mail addresses: yinronghua@mail.kib.ac.cn (R. Yin), xianlikm@163.com (X. Li), zhao.jinhua@yahoo.com (J. Zhao).

https://doi.org/10.1016/j.ijbiomac.2021.07.049
Received 9 April 2021; Received in revised form 13 June 2021; Accepted 6 July 2021
Available online 9 July 2021
0141-8130/© 2021 Elsevier B.V. All rights reserved.

fucose. Some species were also found to produce two distinct FSs. For example, Kariya [15] and Gao [6] reported two kinds of FSs from a sea cucumber, respectively. In a whole, FSs were extensively found in sea cucumbers and their structures vary in a species-specific manner.

The sea cucumber *P. mollis* is a commercially important species in Peru and Chile and it is under commercial exploitation in Mexico, Central and South America [16]. Previously, the polysaccharides have been investigated by our group and a unique FS composed of a repetitively linked backbone →4)-L-Fuc_{2S}-α-(1→ was discovered. In this FS, every trisaccharide unit was branched with Fuc_{4S} and Fuc_{3S} with a molar ratio of 4:1, and it showed strong APTT prolonging activity and intrinsic tenase complex (iXase) inhibitory activity [14]. Recently, it was surprising that five types of FSs were obtained during the reinvestigation on this species. In this work, the purification of the five FSs were described. And their physicochemical properties have been fully characterized by monosaccharides composition, sulfate groups content, molecular weight analysis. The structures of these FSs were analyzed by chemical and instrumental methods including Fourier transform infrared spectroscopy (FT-IR) and 1D/2D nuclear magnetic resonance (NMR) spectroscopy. Among the five kinds of FSs, some are linear molecules consisting of α(1 → 3) or α(1 → 4) glycosidic bonds, respectively. In addition, branched FS was also discovered.

FSs exhibited a wide range of biological activities, among which the anticoagulant activity has attracted wide attention. As a widely-sourced heparin mimics, its development as an anticoagulant has a certain demand. Herein, the anticoagulant activities of the several distinct FSs from *P. mollis* were assayed. The structure and anticoagulant activity analyses were also discussed. These results enriched us the knowledge on structural types of FSs and provided the information to utilize its function.

2. Materials and methods

2.1. Materials

Dried body walls of the sea cucumber *P. mollis* were collected from market in Guangzhou, China. Comparing its partial mitochondrial COI gene sequence with the COI data in BOLDSYSTEMS showed that the sea cucumber was *P. mollis*, with a similarity of 100%. Papain (800 U/mg) was from Shanghai Yuanye Bio-Technology Co., Ltd. (Shanghai, China). Deuterium oxide (99.9%) was from Sigma Aldrich (St. Louis, MI, USA). Amberlite FPA98 ion-exchange resin was purchased from Haas Company (Midland, MI, USA). Sepharose CL-6B, Sephadex G-25 and G-100 were from GE Healthcare. The monosaccharide standards including L-rhamnose (Rha), D-galactose (Gal), D-glucose (Glc), D-glucuronic acid (GlcUA), *N*-acetyl-D-glucosamine (GlcNAc) and 3-methyl-1-phenyl-2-pyrazolin-5-one (PMP) were from Sigma Chemical Co. (St. Louis, MO, USA). D-Mannose (Man), *N*-acetyl-D-galactosamine (GalNAc) and L-fucose (Fuc) were from Aladdin Chemical Reagent Co., Ltd. (Shanghai, China).

Low molecular weight heparin (LMWH) (Enoxaparin, 0.4 mL × 4000 AxaIU) was from Sanofi-Aventis. The activated partial thromboplastin time (APTT), prothrombin time (PT), thrombin time (TT) reagents and normal human plasma were purchased from TECO Medical. All other chemicals were of reagent grade and obtained commercially.

2.2. Extraction and purification of polysaccharides

Crude polysaccharides were isolated from *P. mollis* by the method as previously described with minor modifications [14,17]. Briefly, 2.0 kg dried body wall of *P. mollis* was treated with 0.1% papain aqueous solution at 50 °C for 6 h, followed by treatment with 0.5 M sodium hydroxide at 60 °C for 2 h. After neutralization by adding 6 M HCl, the mixture was centrifuged at 4700 rpm × 15 min to remove the residues. The solution was precipitated by addition of ethanol to a final concentration of 60% and centrifuged at 4700 rpm × 15 min. The precipitate was dissolved in deionized water, the proteins were removed by adjusting the pH to 2.8, staying overnight at 4 °C and centrifuging at 4700 rpm × 15 min. After neutralization, ethanol was added to the obtained supernatant up to a final concentration of 60% (v/v), then centrifuged again. The precipitate was dissolved in deionized water, the solution was decolorized with 3% H_2O_2 at 50 °C for 2 h and centrifuged at 4700 rpm × 15 min. In the present of KOAc (0.5 M), the supernatant was resolved into two fractions by addition of ethanol at the final concentration of 40% (precipitate A) and 60% (v/v) (precipitate B), respectively.

The precipitate B was dissolved in distilled water and further purified by strong anion exchange chromatography using a FPA98 ion-exchange resin column (11.5 × 50 cm), eluted with distilled water, 0.5 M, 1.0 M, 1.5 M, 2.0 M and 3.0 M NaCl solution. The water eluate, 1.5 M NaCl eluate and 3.0 M NaCl eluate was collected, desalted by ultrafiltration with a 1 kDa molecular weight cut-off membrane and freeze-dried to be white powders.

The ^{1}H NMR spectra (Fig. S1) of water and 1.5 M NaCl eluates showed the presence of *N*-acetyl-hexosamine, which can be degraded into oligosaccharide by a partial deacetylation-deamination cleavage method [18]. Therefore, they were subjected to the deacetylation-deamination reaction to remove the trace *N*-acetyl-hexosamine. The water eluate (600 mg) was further purified by gel permeation chromatography (GPC) with Sepharose CL-6B column (2.6 cm × 200 cm) to yield FSI and FSII. The 3.0 M NaCl eluate was further purified by strong anion-exchange (FPA98) chromatography, followed by Sephadex G-100 (1.5 × 150 cm) chromatography to yield FSIII and FSIV. The 1.5 M NaCl eluate was also further purified by FPA98 resin to yield FSV. The purity of each FS was determined by high performance gel permeation chromatography (HPGPC) using Agilent technologies 1200 series (Agilent Co., USA) apparatus with RID (G1361A) and DAD (G1315D) detectors, equipped with a Shodex OHpak SB-804 HQ column (8 × 300 mm). Chromatographic conditions and procedures were performed according to the previous method [19].

2.3. Determination of homogeneity and molecular weight

The molecular mass, including weight-average molecular mass (Mw), number-average molecular mass (Mn) and molecular weight distribution (Mw/Mn) of FSI, FSII, FSIII and FSIV were determined by HPGPC as the previous method [8]. For molecular weight calculation, a standard curve was established by D-series dextran (D-0, 1, 2, 3, 4, 5, 6, 7, 8 and 2000) and was calibrated by an FCS fraction from *H. fuscopunctata* with known molecular weight (Mw 27,760 Da, Mn 24,380 Da) determined by a size-exclusion chromatography with multiple angle laser light scattering on a Shodex OHpak SB-804 HQ column (8 × 300 mm).

2.4. Free radical depolymerization of FSI and FSIII

To elucidate the structures of FSI and FSIII, the low-molecular-weight products were prepared by modified peroxide depolymerization induced by H_2O_2 in the presence of copper (II) [8,20]. In brief, 35 mg and 40 mg of FSI and FSIII was dissolved in distilled water (1.4 mL) and 1.2 mg of copper acetate monohydrate was added at 45 °C, respectively. About 0.7 mL of 30% H_2O_2 was added during 4 h and the crude low-molecular-weight products were precipitated with a final concentration of 80% ethanol (v/v). After centrifugation (8000 rpm × 3 min), desalination on a Sephadex G-25 column (2 × 120 cm), and lyophilization, dFSI and dFSIII was obtained to be white powder with a yield of ~35%.

2.5. General procedures for chemical compositions and physiochemical analysis

Monosaccharide compositions of FSI, FSII, FSIII and FSIV were

determined by HPLC after derivatization with PMP [17].

The sulfate groups contents of FSI, FSII, FSIII and FSIV were determined by a conductimetric method [7]. Specific optical rotations were determined with a Jasco p-1020 polarimeter.

FSI, FSII, FSIII and FSIV were dried under vacuum for 48 h and then pressed with KBr, respectively. The Fourier transform infrared spectroscopy (FT-IR) spectra were collected using a Nicolet iS 10 (Thermo, US) in a range of 400–4000 cm^{-1} at room temperature.

The ^{1}H NMR spectra of FSI, FSII, FSIII and FSIV were recorded in D_2O on a Bruker Avance III 600 MHz spectrometer. The dFSI and dFSIII sample (10 mg) were dissolved in 99.9% D_2O and freeze dried for two times, then they were dissolved in 99.9% D_2O and put into a tube, respectively. ^{1}H, ^{13}C NMR, ^{1}H—^{1}H COSY, TOCSY, ROESY and ^{1}H —^{13}C HSQC, HMBC spectra were recorded using a Bruker Avance III 800 MHz spectrometer. The chemical shifts of ^{1}H were relative to HOD.

2.6. Anticoagulant activity assay

The activated partial thromboplastin time (APTT), prothrombin time (PT) and (thrombin time) TT were determined on a coagulometer (TECOMC-4000, Germany) using APTT, PT and TT reagents and normal human plasma [21].

3. Results and discussion

3.1. Isolation and purification of fucan sulfates

As shown in Fig. 1A, crude polysaccharide was extracted from the dried body wall of *P. mollis* by papain digestion and NaOH treatment. The crude polysaccharides were then fractioned into two fractions, Precipitate A and Precipitate B, by addition of ethanol at the final concentration of 40% and 60% (Fig. 1A). According to our previous research [14], Precipitate A was composed of fucosylated glycosaminoglycan and glucan. The HPGPC profile (Fig. 1B) of Precipitate B showed three peaks with elution time at about 12 min, 14 min and 17 min. Therefore, Precipitate B was further purified by a FPA98 ion-exchange column. The H_2O, 1.5 M NaCl and 3.0 M NaCl eluates were collected and analyzed. The HPGPC profile of the fraction eluted by H_2O showed two peaks at 12 min and 14 min, and the HPGPC profile of the 3.0 M NaCl eluate showed two peaks at 14 min and 17 min, suggesting they were not uniform till this procedure.

In the ^{1}H NMR spectra (Fig. S1) of the components eluted with water, 1.5 M NaCl, the signal at ~2.0 ppm could be assigned to the methyl of *N*-acetyl-hexosamine according to previous studies [14]. The signal at ~1.2 ppm was assigned to the methyl of the fucose residues, and the integral of this signal peak is much greater than that of the signal at ~2.0 ppm, suggesting they were FS components. To further purify these components, they were respectively depolymerized by partial deacetylation-deamination depolymerization [22], a glycosidic bond-selective chemical depolymerization method. After destroying the impurities, the water eluate was purified by Sephadex CL-6B chromatography, according to the Mw difference, to yield two subfractions with retention time at 12 min (FSI) and 14 min (FSII) (Fig. 1C), respectively. The fraction eluted with 1.5 M NaCl was purified by FPA98 with detailed gradient NaCl solution (0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0 M). The HPGPC profile of the fraction eluted with 1.5 M NaCl (FSV) showed a

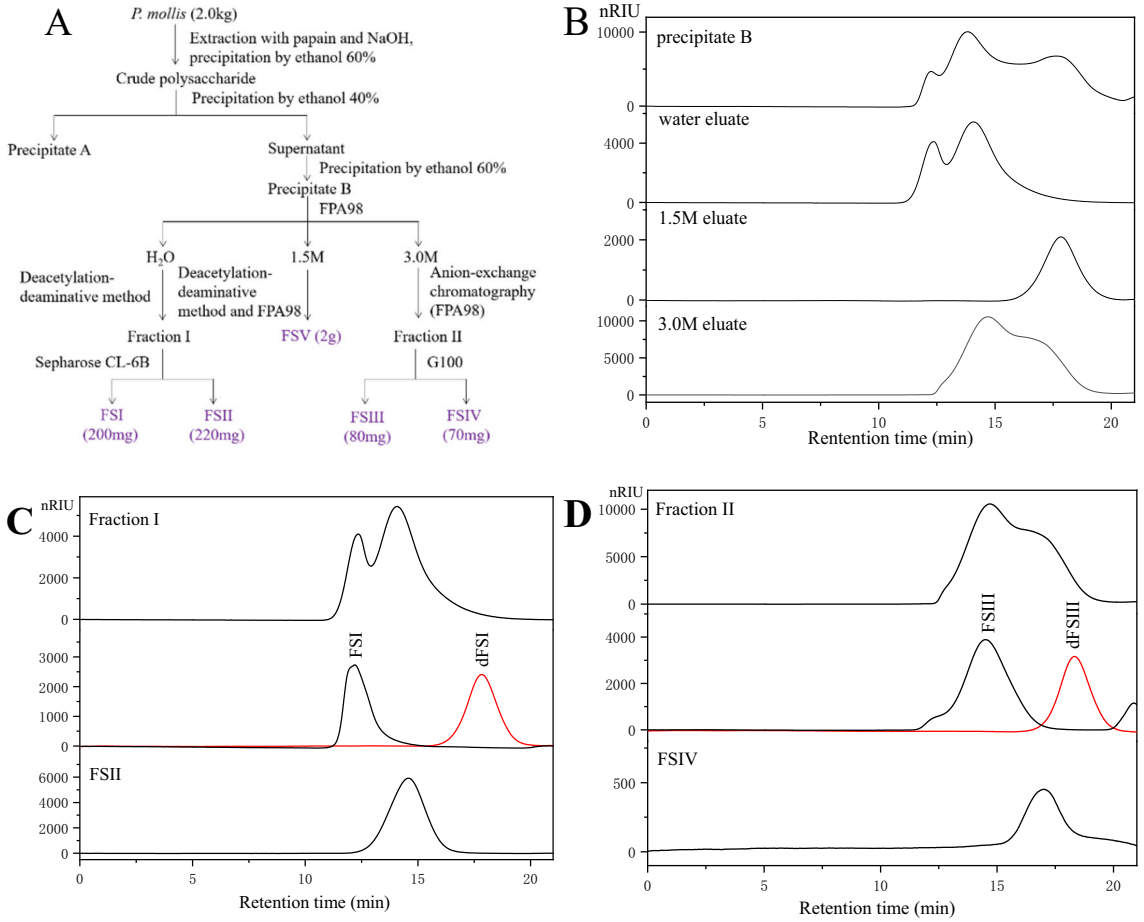


Fig. 1. Flow chart for the purification of FSI, FSII, FSIII and FSIV from the body wall of sea cucumber of *P. mollis* (A) and their HPGPC profiles (B, C, D). The HPGPC profiles of these FSs from the sea cucumber *P. mollis* were determined using a Shodex OHpak SB-804 HQ column (8 mm × 300 mm). Aliquots of 20 μL of samples were analyzed on Agilent Technologies 1200 series apparatus with 0.1 M NaCl as the eluent at a flow rate of 0.5 mL/min.

symmetrical peak in the HPGPC profile (Fig. 1B) and indicated its homogeneity.

The HPGPC profile of the 3.0 M elution showed two peaks, thus it was further fractionated by Sephadex G-100 to yield FSIII with retention time at 14 min and FSIV with retention time at 17 min (Fig. 1D).

Finally, five homogeneous fucan sulfates (FSI, FSII, FSIII, FSIV and FSV) were obtained from the body wall of *P. mollis*. FSV was identified as the reported one (PmFS) in our previous study through its ^{1}H NMR spectrum comparison (Fig. S1) [14]. The structural characteristics of FSI–FSIV were further clarified.

3.2. Chemical composition and physicochemical properties

The physicochemical properties of FSI, FSII, FSIII, FSIV and FSV were measured and compared. As shown in Table 1, FSI had the Mw of 238.2 kDa, similar to other FSs such as that from sea cucumber *H. fuscopunctata* [6]. In addition, the Mw of FSII, FSIII and FSIV was 81.0 kDa, 82.0 kDa and 23.1 kDa, respectively, and the Mw was lower than many other reported FSs as native polysaccharides [2,7,23], which may be related to the life cycle and species [24].

The monosaccharide compositions of FSI, FSII, FSIII and FSIV were shown in Fig. 2A and they were composed of only fucose (Fuc), confirming they were FS components. The sulfate content of FSI, FSII, FSIII and FSIV was measured by a conductimetric method. The conductimetric titration curves (Fig. S2) of FSI, FSII, FSIII and FSIV had only one sharp inflexion point, corresponding to the sulfates. The sulfate contents of FSI, FSII, FSIII and FSIV were calculated to be 32.2%, 29.6%, 28.6% and 35.9%, respectively, and the sulfate ester/fucose monosaccharide molar ratios of FSI, FSII, FSIII and FSIV were 1.00, 0.87, 0.82 and 1.21, respectively (Table 1) similar to those FSs from *H. albiventer* [12], *H. tubolosa* [10], *I. badionotus* [2].

3.3. Structural characterization of FSI, FSII, FSIII and FSIV

The functional groups of the FSs of *P. mollis* were analyzed using IR spectra (Fig. S3). In their IR spectra, the broad signals at 3400–3500 cm^{-1} and the band at 2945 cm^{-1} were characteristics of O—H and C—H stretching vibrations, respectively. The strong peak at 980 cm^{-1} in the fingerprint region was assigned to the asymmetric and symmetric deformation vibrations of the -CH_3 of the fucose residue. The signals at 1020–1070 cm^{-1} were from the stretching vibration of the C-O-C of the polysaccharide skeleton. These signals indicated the existence of a carbohydrate moiety [7,25]. Two signals at 1200–1300 cm^{-1} and 835 cm^{-1} proved the existence of sulfate group; the signal at 1200–1300 cm^{-1} was caused by S═O asymmetric stretching vibration, while that at 835 cm^{-1} was assigned to the symmetric C-O-S stretching vibration of the sulfate. These results confirmed that FSI, FSII, FSIII and FSIV were sulfated polysaccharides. The FT-IR spectra of these FSs was similar to those of fucan sulfates from other species, implying their structural similarity [7,23,26].

The ^{1}H NMR spectra of FSI, FSII, FSIII and FSIV were shown in Fig. 2B. The chemical shift at around δ 1.2 ppm was the methyl protons of L-fucose residue. FSI, FSII and FSIII gave overlapped ^{1}H NMR spectra with broad signals, which might be attributed to its high Mw. FSI and FSII showed the same signals in the ^{1}H NMR spectra. To elucidate the structures in detail, the low-molecular-weight depolymerized products (dFSI, dFSIII) were prepared from FSI and FSIII by the free radical depolymerization method. The resolution of the ^{1}H NMR spectra of dFSI and dFSIII was significantly enhanced (Figs. 3 and 4, and S4).

The ^{1}H/^{13}C NMR spectra (Fig. 3A) of dFSI showed complex and challenging resolution. The strong signals at around 1.1–1.3 ppm in the ^{1}H NMR spectrum were the methyl protons from the L-fucose residue, and the signal observed at 4.93–5.03 ppm, 5.02–5.15 ppm, 5.15–5.36 ppm, 5.38–5.50 ppm could be assigned to the anomeric protons of four α-L-fucose residues with different sulfate positions. The ^{1}H—^{1}H COSY and TOCSY spectra were used to assign the proton signals of the four residues (Fig. 3B, Table 2). Carbon chemical shifts of the residues were also assigned using the ^{1}H—^{13}C HSQC spectrum (Fig. 3C). The complete assignment of dFSI is shown in Table 2. The sulfated positions of each fucose residue were further elucidated by comparing the chemical shift of each proton with that of non-sulfated fucose. Generally, the H chemical shift of sulfated positions in fucose residue would shifted downfield by about 0.4–0.8 ppm [27]. Therefore, according to full chemical shifts assignments, the different sulfated α-L-Fuc residues were clarified as Fuc_{0S}, Fuc_{4S}, Fuc_{2S} and Fuc_{2S4S}, respectively. The glycosidic linkages were clarified according to the correlation peaks in ^{1}H—^{1}H ROESY spectrum (Fig. 3B). The correlations between H-1 and H-3 of these residues result from the α (1 → 3) linkages, which is the most prevalent glycosidic types found in the sea cucumber FS. Overall, the primary structure of dFSI was elucidated and it is a linear fucan sulfate composed of →3)-α-L-Fuc_S-(1→. The sulfated types of Fuc contain Fuc_{0S}, Fuc_{4S}, Fuc_{2S} and Fuc_{2S4S} with the ratio of about 3:3:2:2 (Fig. 3D).

According to the ^{1}H NMR spectra (Fig. S4) of FSI and dFSI, the chemical shifts of the characteristic peaks of FSI and dFSI remained the same, while the proportion of the peak area of the anomeric protons from different sulfated fucose have changed, especially that of Fuc_{2S} decreased significantly. This may be caused by the loss of Fuc_{2S} during the aftertreatment of the peroxide depolymerization. Combing the ^{1}H NMR data and the sulfate ester/fucose monosaccharide molar ratio calculated by the conductimetric method, native FSI was proven as →3)-L-FucS-α(1→, where FucS were Fuc_{0S} (10%), Fuc_{4S} (36%), Fuc_{2S} (44%) and Fuc_{2S4S} (10%).

^{1}H NMR spectra of FSI and FSII showed that FSII has the same primary structure with FSI. That is, FSII is also a linear polysaccharide formed by FucS connected by (α1 → 3) glycosidic bonds. Whereas, the molecular weight of FSII is about three times lower than that of FSI, indicating that the chain length of FSII is two times shorter than that of FSI.

In the ^{1}H NMR spectrum of dFSIII (Fig. 4A), three main signals at 5.09–5.19 ppm, 5.18–5.48 ppm, 4.91–5.09 ppm were observed and

Table 1
Chemical composition, physicochemical properties, anticoagulant activity of FSI, FSII, FSIII, FSIV and FSV from the sea cucumber of *P. mollis*.

Sulfated polysaccharide	Average molecular weight[a]			Specific rotation [α]	Chemical composition (molar ratio)			-OSO_3^- (%)	APTT[c]	PT[c]	TT[c]
	Mw (kDa)	Mn (kDa)	Mw/Mn PDI		GlcUA	GalNAc	Fuc		(μg/mL)	(μg/mL)	(μg/mL)
FSI	238.3	117.5	2.02	−102.5	0	0	1	32.2	11.81	>128[d]	40.60
FSII	81.0	32.6	2.48	−153.4	0	0	1	29.6	20.22	>128	110.40
FSIII	82.0	35.9	2.28	−118.0	0	0	1	28.6	23.34	>128	121.90
FSIV	23.1	11.3	2.05	−128.5	0	0	1	35.9	23.59	>128	>128
FSV[b]	6.12	ND	ND	−115.2	0	0	1	30.4	24.3	>128	>128
LMWH	~4.5	ND	ND	ND	–	–	–	ND	7.52	>128	6.24

[a] Molecular weights were determined by HPGPC methods ((Materials and Methods).
[b] Data from ref. [14].
[c] Concentration required to double the APTT or PT or TT of human plasma.
[d] >128, this activity was not significant at concentration up to 128 μg/mL.

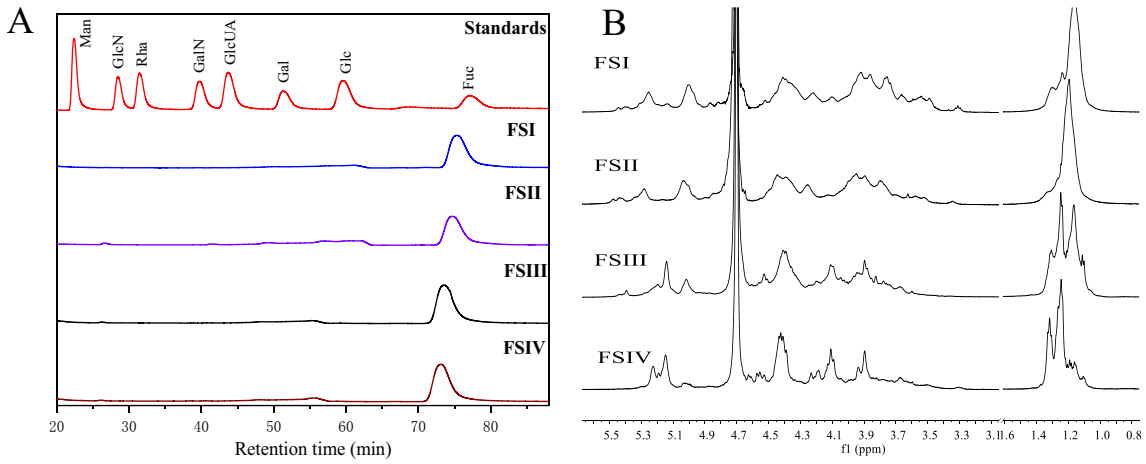


Fig. 2. HPLC profiles of PMP derivatives of mixed monosaccharide standards, FSI, FSII, FSIII and FSIV (A); ^{1}H NMR spectra of FSI, FSII, FSIII and FSIV (B).

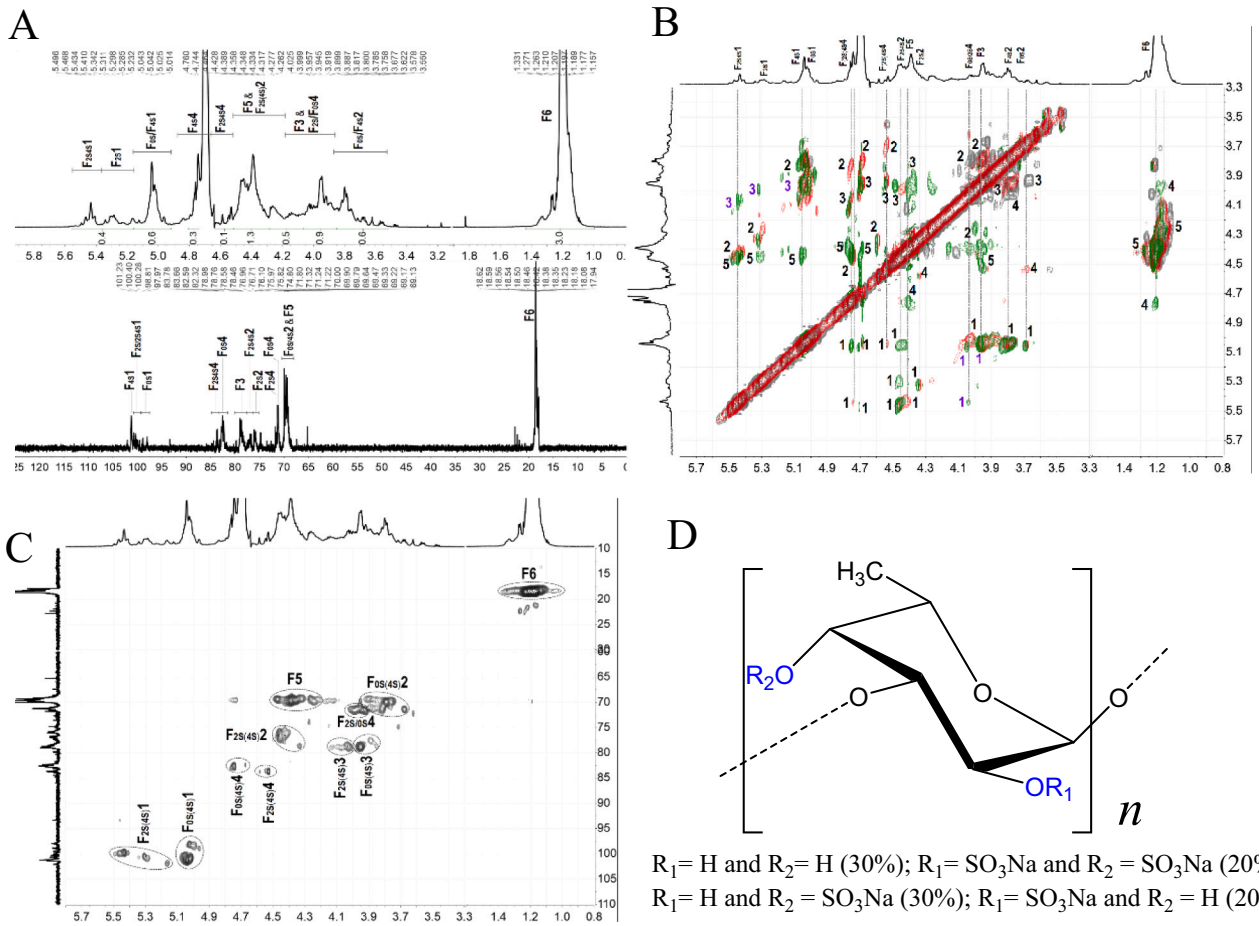


Fig. 3. ^{1}H/^{13}C NMR (A) of dFSI; Superimposed spectra of ^{1}H—^{1}H COSY (gray), TOCSY (red), ROESY (green) (B), ^{1}H—^{13}C HSQC (C) and its chemical structure (D). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

could be assigned to the anomeric protons of three α-L-fucose residues A, B, C. According to the integral area of these anomeric protons, the ratio of A: B: C was about 1:1:1. To assign other proton signals, ^{1}H—^{1}H COSY and ^{1}H—^{1}H TOCSY spectra were employed to complete these protons assignment (Fig. 4B, Table 3). Additionally, the chemical shifts values of all carbons in dFSIII were fully assigned according to the ^{1}H—^{13}C HSQC and presented in Table 3. Furthermore, structural features including the glycosidic linkages and residue sequences were confirmed according to the correlation peaks in ^{1}H—^{1}H ROESY (Fig. 4B) and ^{1}H—^{13}C HMBC spectra (Fig. S5). The correlation between H-1 and H-4 of these residues was identified in the ^{1}H—^{1}H ROESY spectrum and the long-range correlations in the ^{1}H—^{13}C HMBC spectrum, confirming the presence of 1 → 4 linkages in FSIII. Comparing the chemical shift of each proton with that of non-sulfated fucose residue [28,29], both A and B could be deduced to Fuc_{2S}. The O-3 position of residue C was sulfated (Fuc_{3S}) in about 50%. In Fig. 4A, B_1' was also assigned as the anomeric signal of

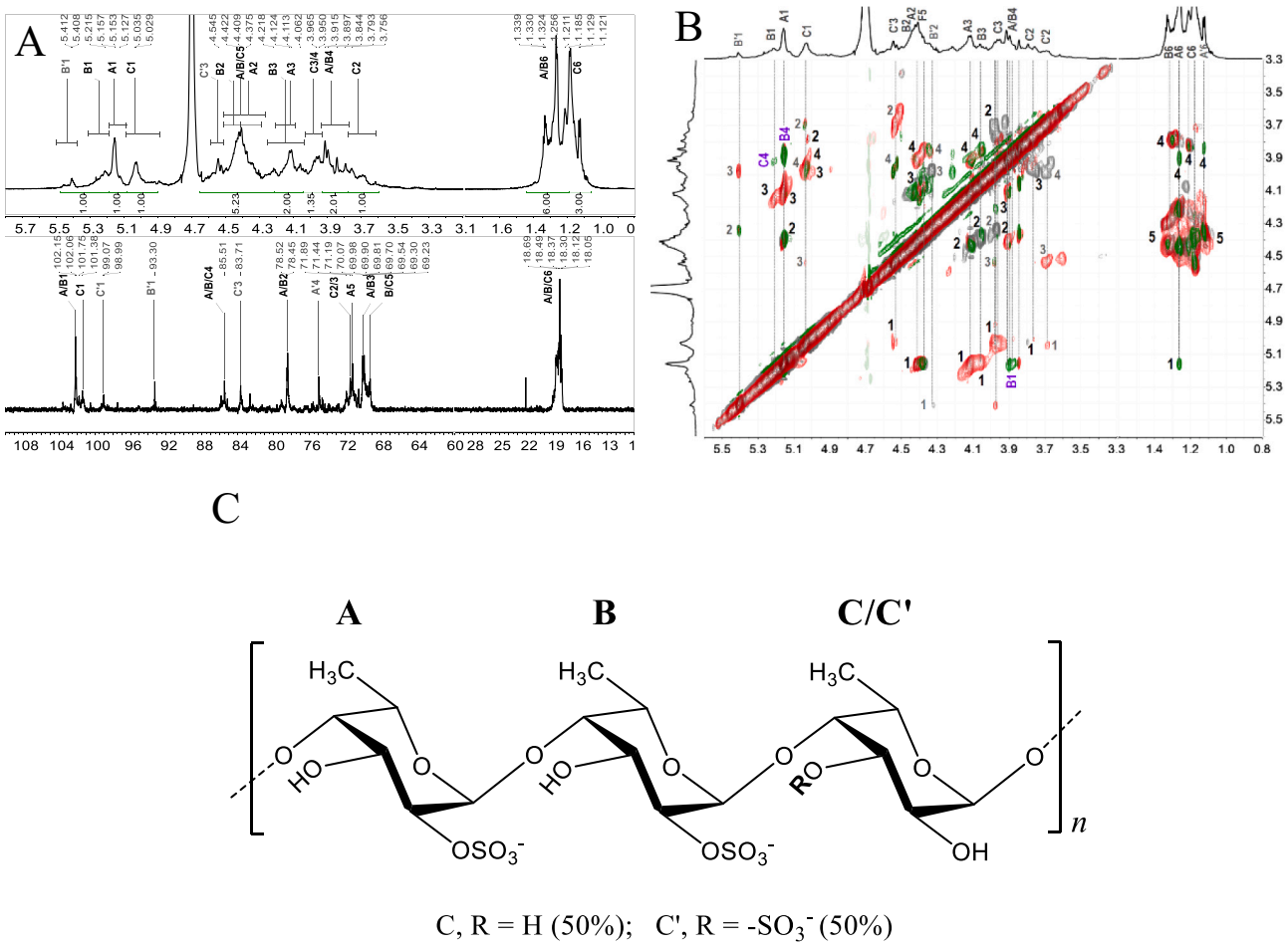


Fig. 4. $^{1}H/^{13}C$ NMR (A) of dFSIII; Superimposed spectra of $^{1}H—^{1}H$ COSY (gray), TOCSY (red), ROESY (green) (B) and its chemical structure (C). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2
^{1}H and ^{13}C NMR chemical shifts assignments of dFSI.

Residues		Chemical shifts (ppm)					
		1	2	3	4	5	6
Fuc_{0S}	^{1}H	4.93–5.03	3.65–3.85	3.90–3.99	3.90–4.05	4.19–4.52	1.08–1.28
	^{13}C	97.53–99.16	69.50–70.20	78.05–79.88	70.83–71.36	68.38–70.20	17.88–18.98
Fuc_{4S}	^{1}H	5.02–5.15	3.72–3.90	3.90–3.99	4.68–4.82	4.19–4.52	1.08–1.28
	^{13}C	100.95–101.50	69.50–70.20	78.05–79.88	81.73–83.09	68.38–70.20	17.88–18.98
Fuc_{2S}	^{1}H	5.15–5.36	4.29–4.40	3.99–4.19	3.90–4.05	4.19–4.52	1.08–1.28
	^{13}C	99.35–100.90	75.50–76.30	78.05–79.88	71.49–71.93	68.38–70.20	17.88–18.98
Fuc_{2S4S}	^{1}H	5.38–5.50	4.42–4.50	3.99–4.19	4.50–4.60	4.19–4.52	1.08–1.28
	^{13}C	99.35–100.90	76.60–77.60	78.05–79.88	83.51–84.80	68.38–70.20	17.88–18.98

Table 3
^{1}H and ^{13}C NMR chemical shifts assignments of dFSIII.

Residues			Chemical shifts (ppm)					
			1	2	3	4	5	6
A	→4-L-Fuc_{2S}-α1→	^{1}H	5.09–5.19	4.31–4.39	4.09–4.17	3.88–3.93	4.39–4.49	1.23–1.30
		^{13}C	101.8–102.3	78.0–79.5	69.8–70.3	85.4–85.6	70.9–71.3	17.5–19.5
B	→4-L-Fuc_{2S}-α1→	^{1}H	5.18–5.48	4.37–4.47	4.03–4.26	3.85–3.93	4.32–4.36	1.20–1.38
		^{13}C	101.6–102.3	78.0–79.5	69.8–70.3	85.4–86.1	68.9–69.8	17.5–19.5
C/C′	→4-L-Fuc_{0S}-α1 → or →4-L-Fuc_{3S}-α1→	^{1}H	4.91–5.09	3.66–3.78 (3.74–3.82)[a]	3.93–4.03 (4.52–4.60)[a]	3.91–3.97	4.32–4.36	1.08–1.20
		^{13}C	101.1–101.6 (98.7–99.4)[a]	71.3–72.6	71.6–72.3 (83.4–83.8)[a]	85.0–85.4	68.9–69.8	17.5–19.5

[a] ^{1}H and ^{13}C NMR chemical shifts of C′, C′ was Fuc_{3S}.

Fuc_{2S}, according to the correlations in Fig. 4B. While its anomeric carbon (93.3 ppm) indicated the Fuc_{2S} residue was located at the reducing end in the chain of dFSIII, which may be attributed to the preferential cleavage here. Overall, combination of chemical analysis allowed us to determine the structure of the FSIII and it was deduced as →4)-L-Fuc_{2S}-(α1 → 4)-L-Fuc_{2S}-(α1 → 4)-L-$Fuc_{0S/3S}$-(α1→, Fuc_{0S} and Fuc_{3S} were 50% in molar, respectively.

The structure of FSIV was also elucidated by the detailed analysis of its 1D/2D NMR spectra. In the ^1H NMR spectrum, the signals at 5.22 ppm, 5.23 ppm, 5.18 ppm, 5.15 ppm and 5.16 ppm could be assigned to the anomeric protons of α-L-fucose residues A_I, A_{II}, B_I, B_{II} and B_{III}, respectively. Correspondingly, the other H signals could be found from the ^1H—^1H TOCSY spectrum (Fig. 5B) starting from them. The carbon signals could be further assigned according to the ^1H—^{13}C HSQC spectrum (Fig. S6) and was summarized in Table 4. The sulfation positions of each fucose residue were further elucidated by their downfield-shifted chemical signals. Residue A_I and A_{II} was deduced to be 2,3-di-O-sulfated fucose residue as H-2 (4.555 ppm in A_I, 4.702 ppm in A_{II}) and H-3 (4.619 ppm in A_I, 4.712 ppm in A_{II}) signals apparently shifted to downfield. In the same way, residue B_I, B_{II} and B_{III} was clarified to be 2-O-sulfated fucose residue. The glycosidic linkage and the sequence of A_I, A_{II}, B_I, B_{II} and B_{III} residues were clarified according to the correlation peaks in ^1H—^1H ROESY spectrum (Fig. 5B). In detail, correlation from H-1 of residue A_I to H-4 of residue A_{II}, H-1 of residue A_{II} to H-4 of residue B_I, H-1 of residue B_I to H-4 of residue B_{II}, H-1 of residue B_{II} to H-4 of residue B_{III}, H-1 of residue B_{III} to H-4 of residue A_I, respectively. Then the sequence of these five residues in FSV could be deduced as →4A_Iα1 → 4A_{II}α1 → 4B_Iα1 → 4B_{II}α1 → 4B_{III}α1→, indicating they were linked in alternating α1 → 4 linkages. Moreover, NMR analysis indicates that A_I, A_{II}, B_I and B_{III} are present in equal proportions, and B_{II} is present in twice that proportion. Finally, the structure of FSIV was deduced as →4)-L-Fuc_{2S3S}-(α1 → 4)-L-Fuc_{2S3S}-(α1 → 4)-L-Fuc_{2S}-(α1→4)-L-Fuc_{2S}-(α1 → 4)-L-Fuc_{2S}-(α1 → 4)-L-Fuc_{2S}-(α1 → .

At present, we have discovered five types of FS from sea cucumbers *P. mollis*. The types of glycosidic bonds include (α1 → 3) and (α1 → 4). Sulfated substitutions occur at positions O-2, O-3, or O-4, but ratio of different sulfated substituted fucan and the order of connection are rich and diverse, which further enriches the diversity of structural types of FS derived from sea cucumbers. And this is the first time to obtain so many types of FS in one sea cucumber species, indicating the structure diversity of FS in sea cucumber.

3.4. Anticoagulant activities of FSI, FSII, FSIII and FSIV in vitro

Sulfated polysaccharides found in marine organisms usually have significant anticoagulant and antithrombotic properties [30,31]. Because biological property depends on their structural characteristics, the structural features variation of FSI, FSII, FSIII and FSIV prompted us to investigate the anticoagulant activities by detecting their effects on the APTT, PT, and TT of normal human plasma, compared with that of LMWH. Such assays are routinely used to determine the ability of anticoagulants to inhibit blood clotting through the intrinsic, extrinsic and common pathways of the coagulation cascade, respectively [32,33]. The results showed that these four sulfated polysaccharides showed a prolongation of clotting by APTT in a concentration-dependent manner (Fig. 6 and Table 1).

The concentrations of FSI, FSII, FSIII and FSIV required to double the APTT were 11.81, 20.22, 23.34 and 23.59 μg/mL, respectively. Although their concentrations required for double the APTT were about 2 to 3-fold of that of LMWH, they also had significant anticoagulant activity. In the TT assays, FSI, FSII and FSIII exhibited weak activity and the concentration required to double TT were 40.60, 110.40, 121.90 μg/mL, respectively, while FSIV exhibited no effect on TT at the concentration up to 128 μg/mL. The results indicated that the four FS exhibited potent anticoagulant activities by targeting the intrinsic pathway. No obvious effect on PT of human plasma at the concentrations up to 128 μg/mL indicated that they had no effect on the extrinsic coagulation pathways, which agreed with the anticoagulant activity evaluation of other fucan sulfates [6,7]. The APTT and TT prolonging activities of FSI were more potent than those of FSII~IV. Considering that FSI and FSII possessed the same structural sequence and sulfation patterns with different Mws. The discrepancy in their anticoagulant activity could be mainly due to the molecular weight. For FSII and FSIII, they possessed the similar Mw and degree of sulfation, while linked by different glycosidic bonds (α1,3 *vs* α1,4). This suggested that glycosidic bond was not the key factor for the activity. For FSIII and FSIV, the two owned the

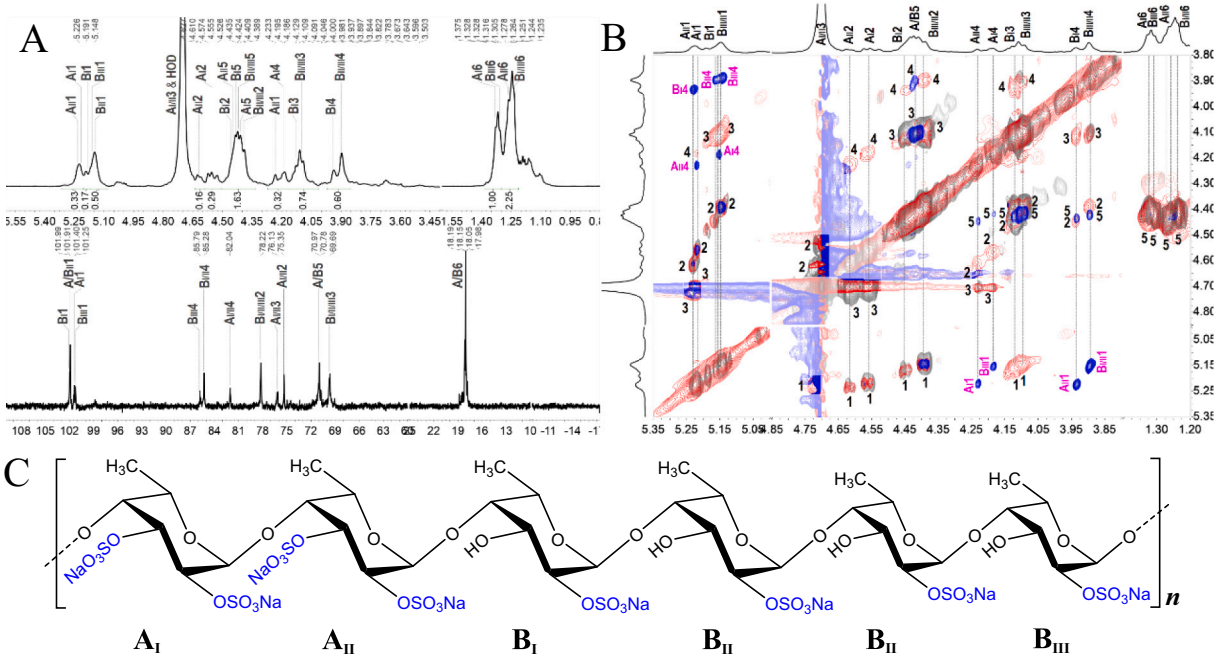


Fig. 5. $^1H/^{13}C$ NMR (A) of FSIV; Superimposed spectra of ^1H—^1H COSY (gray), TOCSY (red), ROESY (blue) (B) and its chemical structure (C). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 4
^{1}H and ^{13}C NMR chemical shifts assignments of FSIV.

Residues			Chemical shifts (ppm)					
			1	2	3	4	5	6
A_I	→4-L-Fuc_{2S3S}-α1→	^{1}H	5.218	4.555	4.702	4.191	4.409	1.325
		^{13}C	101.40	75.35	76.13	82.04	70.78	18.19
A_{II}	→4-L-Fuc_{2S3S}-α1→	^{1}H	5.230	4.619	4.712	4.233	4.446	1.270
		^{13}C	101.91	75.35	76.13	82.04	70.97	18.05
B_I	→4-L-Fuc_{2S}-α1→	^{1}H	5.182	4.458	4.129	3.937	4.432	1.244
		^{13}C	101.99	78.22	69.69	85.28	70.97	17.98
B_{II}	→4-L-Fuc_{2S}-α1→	^{1}H	5.146	4.396	4.109	3.897	4.419	1.244
		^{13}C	101.91	78.22	69.69	85.28	70.97	18.05
B_{III}	→4-L-Fuc_{2S}-α1→	^{1}H	5.160	4.396	4.109	3.897	4.419	1.309
		^{13}C	101.25	78.22	69.69	85.79	70.78	18.16

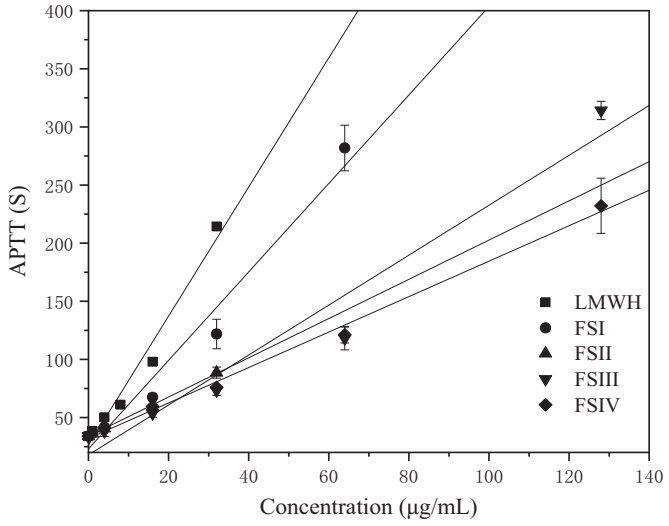


Fig. 6. APTT prolonging activities of FSI, FSII, FSIII and FSIV on APTT in normal human plasma.

same linkages (α1,4), while FSIV had higher sulfation degree and lower Mw, indicating that sulfation degree is another crucial element for anticoagulation.

4. Conclusion

In this study, five fucan sulfates (FSI, FSII, FSIII, FSIV and FSV) were purified from a sea cucumber species (*P. mollis*) and their structures were characterized by physicochemical analysis and NMR spectra. The primary structure of FSI was clarified as →3)-L-FucS-α(1→, where FucS were Fuc_{0S} (10%), Fuc_{4S} (36%), Fuc_{2S} (44%) and Fuc_{2S4S} (10%). FSII has the same structure feature with FSI, while with lower molecular mass. The structure of FSIII was clarified as the trisaccharide repeating unit →4)-L-Fuc_{2S}-(α1 → 4)-L-Fuc_{2S}-(α1 → 4)-L-$Fuc_{0S/3S}$-(α1→, where Fuc_{0S} and Fuc_{3S} were 50%, respectively. The repeating unit of FSIV, →4)-L-Fuc_{2S3S}-(α1 → 4)-L-Fuc_{2S3S}-(α1 → 4)-L-Fuc_{2S}-(α1→4)-L-Fuc_{2S}-(α1 → 4)-L-Fuc_{2S}-(α1 → 4)-L-Fuc_{2S}-(α1 → , was hexasaccharide organized by the distinctive (α1 → 4) glycosidic bonds. With FSV, its backbone consisted of a repetitively linked unit {→4-L-Fuc_{2S}-α1→}, and every trisaccharide unit was branched with Fuc_{4S} or Fuc_{3S} in a molar ratio of 4:1. The glycosidic linkages of FSIII, FSIV and FSV are different from that of FSI and FSII. Till now, this is the first report that such a variety of fucan sulfate have been obtained from the same sea cucumber species. The structure showed distinctly different characteristics, branched or linear ones. Among the linear FSs, the structural variations embodied in the glycosidic bonds, number of the Fuc residues in every repeated unit, and the sulfation patterns. Such structural diversity leads to an inquiry into the physiological function, and the biosynthesis mechanism. Additionally, anticoagulant activity assays indicated that FSI, FSII, FSIII and FSIV showed strong APTT prolonging activities, and the activity discrepancy should result from the difference in Mw and sulfation degree. This also may provide more information for the research of anticoagulant drugs.

CRediT authorship contribution statement

Jinhua Zhao, Na Gao and Xian Li conceived the idea, designed the experiments, supervised students to carry out the experiments, revised the paper. Yan Ma, Zhichuang Zuo, Shanni Li, Wenqi Zheng, Xiang Shi, Qipei Liu and Ting Ma performed all the tests. Yan Ma and Ronghua Yin wrote and revised the manuscript.

Declaration of competing interest

The authors state no conflict of interest.

Acknowledgments

The work is supported in part by the National Natural Science Foundation of China (82073725, 81703374, and 81773737).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijbiomac.2021.07.049.

References

[1] Q. Han, J.K. Keesing, D. Liu, A review of sea cucumber aquaculture, ranching, and stock enhancement in China, Rev. Fish. Sci. Aquac. 24 (4) (2016) 326–341.
[2] S. Chen, Y. Hu, X. Ye, G. Li, G. Yu, C. Xue, W. Chai, Sequence determination and anticoagulant and antithrombotic activities of a novel fucan sulfate isolated from the sea cucumber isostichopus badionotus, Biochim. Biophys. Acta 1820 (7) (2012) 989–1000.
[3] C. Li, Q. Niu, S. Li, X. Zhang, C. Liu, C. Cai, G. Li, G. Yu, Fucoidan from sea cucumber Holothuria polii: structural elucidation and stimulation of hematopoietic activity, Int. J. Biol. Macromol. 154 (2020) 1123–1131.
[4] P.D. Thinh, B.M. Ly, R.V. Usoltseva, N.M. Shevchenko, A.B. Rasin, S.D. Anastyuk, O.S. Malyarenko, T.N. Zvyagintseva, P.T. San, S.P. Ermakova, A novel fucan sulfate from vietnamese sea cucumber Stichopus variegatus: isolation, structure and anticancer activity in vitro, Int. J. Biol. Macromol. 117 (2018) 1101–1109.
[5] J. Wang, S. Hu, W. Jiang, W. Song, L. Cai, J. Wang, Fucoidan from sea cucumber may improve hepatic inflammatory response and insulin resistance in mice, Int. Immunopharmacol. 31 (2016) 15–23.
[6] N. Gao, R. Chen, R. Mou, J. Xiang, K. Zhou, Z. Li, J. Zhao, Purification, structural characterization and anticoagulant activities of four sulfated polysaccharides from sea cucumber Holothuria fuscopunctata, Int. J. Biol. Macromol. 164 (2020) 3421–3428.
[7] X. Li, S. Li, J. Liu, L. Lin, H. Sun, W. Yang, Y. Cai, N. Gao, L. Zhou, H. Qin, R. Yin, J. Zhao, A regular fucan sulfate from Stichopus herrmanni and its peroxide depolymerization: structure and anticoagulant activity, Carbohydr. Polym. 256 (2021), 117513.
[8] F. Shang, R. Mou, Z. Zhang, N. Gao, L. Lin, Z. Li, M. Wu, J. Zhao, Structural analysis and anticoagulant activities of three highly regular fucan sulfates as novel intrinsic factor xase inhibitors, Carbohydr. Polym. 195 (2018) 257–266.

[9] L. Yu, L. Ge, C. Xue, Y. Chang, C. Zhang, X. Xu, Y. Wang, Structural study of fucoidan from sea cucumber acaudina molpadioides: a fucoidan containing novel tetrafucose repeating unit, Food Chem. 142 (2014) 197–200.
[10] Y. Chang, Y. Hu, L. Yu, D.J. McClements, X. Xu, G. Liu, C. Xue, Primary structure and chain conformation of fucoidan extracted from sea cucumber Holothuria tubulosa, Carbohydr. Polym. 136 (2016) 1091–1097.
[11] L. Yu, C. Xue, Y. Chang, X. Xu, L. Ge, G. Liu, Y. Wang, Structure elucidation of fucoidan composed of a novel tetrafucose repeating unit from sea cucumber thelenota ananas, Food Chem. 146 (2014) 113–119.
[12] Y. Cai, W. Yang, R. Yin, L. Zhou, Z. Li, M. Wu, J. Zhao, An anticoagulant fucan sulfate with hexasaccharide repeating units from the sea cucumber Holothuria albiventer, Carbohydr. Res. 464 (2018) 12–18.
[13] L. Yu, C. Xue, Y. Chang, Y. Hu, X. Xu, L. Ge, G. Liu, Structure and rheological characteristics of fucoidan from sea cucumber Apostichopus japonicus, Food Chem. 180 (2015) 71–76.
[14] W. Zheng, L. Zhou, L. Lin, Y. Cai, H. Sun, L. Zhao, N. Gao, R. Yin, J. Zhao, Physicochemical characteristics and anticoagulant activities of the polysaccharides from sea cucumber pattalus mollis, Mar. Drugs. 17 (4) (2019) 198.
[15] Y. Kariya, B. Mulloy, K. Imai, A. Tominaga, T. Kaneko, A. Asari, K. Suzuki, H. Masuda, M. Kyogashima, T. Ishii, Isolation and partial characterization of fucan sulfates from the body wall of sea cucumber Stichopus japonicus and their ability to inhibit osteoclastogenesis, Carbohydr. Res. 339 (7) (2004) 1339–1346.
[16] E. Garcia-Candela, C. Pariona-Velarde, A. Mondragón-Martínez, V. Chumpitaz-Cerrate, Antiviral activity of the sea cucumber tegument extract (Pattalus mollis) on human rotavirus a (RVA), Nat. Prod. Res. 35 (6) (2019) 1014–1018.
[17] X. Li, L. Luo, Y. Cai, W. Yang, L. Lin, Z. Li, N. Gao, S.W. Purcell, M. Wu, J. Zhao, Structural elucidation and biological activity of a highly regular fucosylated glycosaminoglycan from the edible sea cucumber Stichopus herrmanni, J. Agric. Food Chem. 65 (42) (2017) 9315–9323.
[18] L. Zhao, S. Lai, R. Huang, M. Wu, N. Gao, L. Xu, H. Qin, W. Peng, J. Zhao, Structure and anticoagulant activity of fucosylated glycosaminoglycan degraded by deaminative cleavage, Carbohydr. Polym. 98 (2) (2013) 1514–1523.
[19] L. Luo, M. Wu, L. Xu, W. Lian, J. Xiang, F. Lu, N. Gao, C. Xiao, S. Wang, J. Zhao, Comparison of physicochemical characteristics and anticoagulant activities of polysaccharides from three sea cucumbers, Mar. Drugs. 11 (2) (2013) 399–417.
[20] M. Wu, S. Xu, J. Zhao, H. Kang, H. Ding, Free-radical depolymerization of glycosaminoglycan from sea cucumber thelenata ananas by hydrogen peroxide and copper ions, Carbohydr. Polym. 80 (4) (2010) 1116–1124.
[21] M. Wu, D. Wen, N. Gao, C. Xiao, L. Yang, L. Xu, W. Lian, W. Peng, J. Jiang, J. Zhao, Anticoagulant and antithrombotic evaluation of native fucosylated chondroitin sulfates and their derivatives as selective inhibitors of intrinsic factor xase, Eur. J. Med. Chem. 92 (2015) 257–269.
[22] L. Zhao, S. Lai, R. Huang, M. Wu, N. Gao, L. Xu, H. Qin, W. Peng, J. Zhao, Structure and anticoagulant activity of fucosylated glycosaminoglycan degraded by deaminative cleavage, Carbohydr. Polym. 98 (2) (2013) 1514–1523.
[23] W. He, H. Sun, L. Su, D. Zhou, X. Zhang, D. Shanggui, Y. Chen, Structure and anticoagulant activity of a fucan sulfate from the sea cucumber acaudina leucoprocta, Int. J. Biol. Macromol. 164 (2020) 87–94.
[24] J. Feng, L. Zhang, X. Tang, X. Xia, W. Hu, P. Zhou, Season and geography induced variation in sea cucumber (Stichopus japonicus) nutritional composition and gut microbiota, J. Food Compos. Anal. 103838 (2021).
[25] B. Matsuhiro, I.O. Osorio-Román, R. Torres, Vibrational spectroscopy characterization and anticoagulant activity of a sulfated polysaccharide from sea cucumber athyonidium chilensis, Carbohydr. Polym. 88 (3) (2012) 959–965.
[26] M. Wu, L. Xu, L. Zhao, C. Xiao, N. Gao, L. Luo, L. Yang, Z. Li, L. Chen, J. Zhao, Structural analysis and anticoagulant activities of the novel fucan sulfate possessing a regular well-defined repeating unit from sea cucumber, Mar. Drugs. 13 (4) (2015) 2063–2084.
[27] S. Yamada, K. Yoshida, M. Sugiura, K. Sugahara, One-and two-dimensional 1H-NMR characterization of two series of sulfated disaccharides prepared from chondroitin sulfate and heparan sulfate/heparin by bacterial eliminase digestion, J. Biochem. 112 (4) (1992) 440–447.
[28] P.K. Agrawal, NMR spectroscopy in the structural elucidation of oligosaccharides and glycosides, Phytochemistry 31 (10) (1992) 3307–3330.
[29] W. Yang, Y. Cai, R. Yin, L. Lin, Z. Li, M. Wu, J. Zhao, Structural analysis and anticoagulant activities of two sulfated polysaccharides from the sea cucumber holothuria coluber, Int. J. Biol. Macromol. 115 (2018) 1055–1062.
[30] H. Li, Q. Yuan, K. Lv, H. Ma, C. Gao, Y. Liu, S. Zhang, L. Zhao, Low-molecular-weight fucosylated glycosaminoglycan and its oligosaccharides from sea cucumber as novel anticoagulants: a review, Carbohydr. Polym. 251 (2021), 117034.
[31] C. Xu, R. Zhang, Z. Wen, Bioactive compounds and biological functions of sea cucumbers as potential functional foods, J. Funct. Food. 49 (2018) 73–84.
[32] R.A. Al-Horani, P. Ponnusamy, A.Y. Mehta, D. Gailani, U.R. Desai, Sulfated pentagalloylglucoside is a potent, allosteric, and selective inhibitor of factor XIa, J. Med. Chem. 56 (3) (2013) 867–878.
[33] F. Shang, N. Gao, R. Yin, L. Lin, C. Xiao, L. Zhou, Z. Li, S.W. Purcell, M. Wu, J. Zhao, Precise structures of fucosylated glycosaminoglycan and its oligosaccharides as novel intrinsic factor xase inhibitors, Eur. J. Med. Chem. 148 (2018) 423–435.